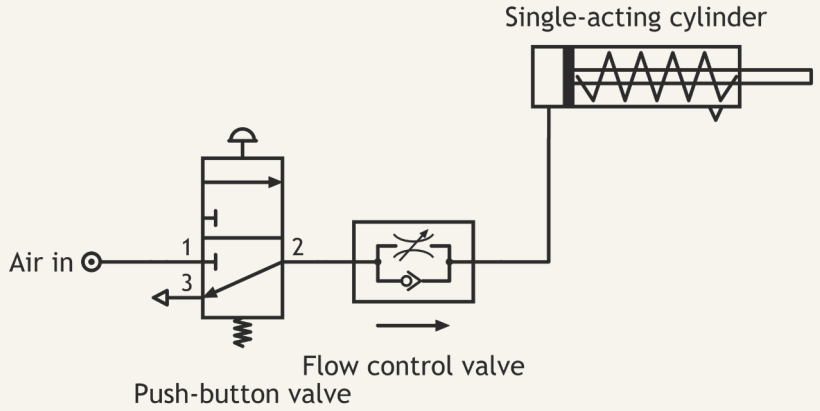
Daily Review

1	How many states has a 3/2 valve?	2
2	What does a T in a box mean?	That port is blocked
3	What brings a single-acting rod back in?	Its spring
4	The middle box of a system diagram?	Process
5	The unit of force?	The newton (N)

Control the Flow

How do you stop the gate pin slamming?



S3 Engineering Science

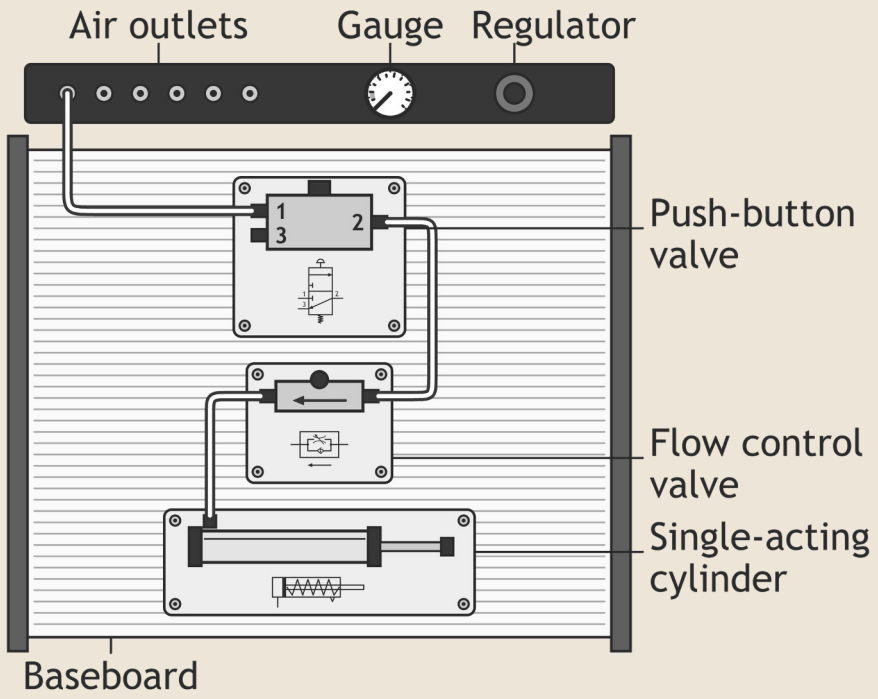
The Job: Stop the Slam

i When the button is pressed, the rod moves out slowly.

ii When the screw is turned in, the rod moves out more slowly.

iii When the button is let go, the rod moves back in.

Predict: does it slow the rod both ways?



By the end of this lesson…



1. I can slow a cylinder with a flow control valve
2. I can plan a fair test: keep the same, change one thing
3. I can say which way the arrow slows the air

THE BUILD

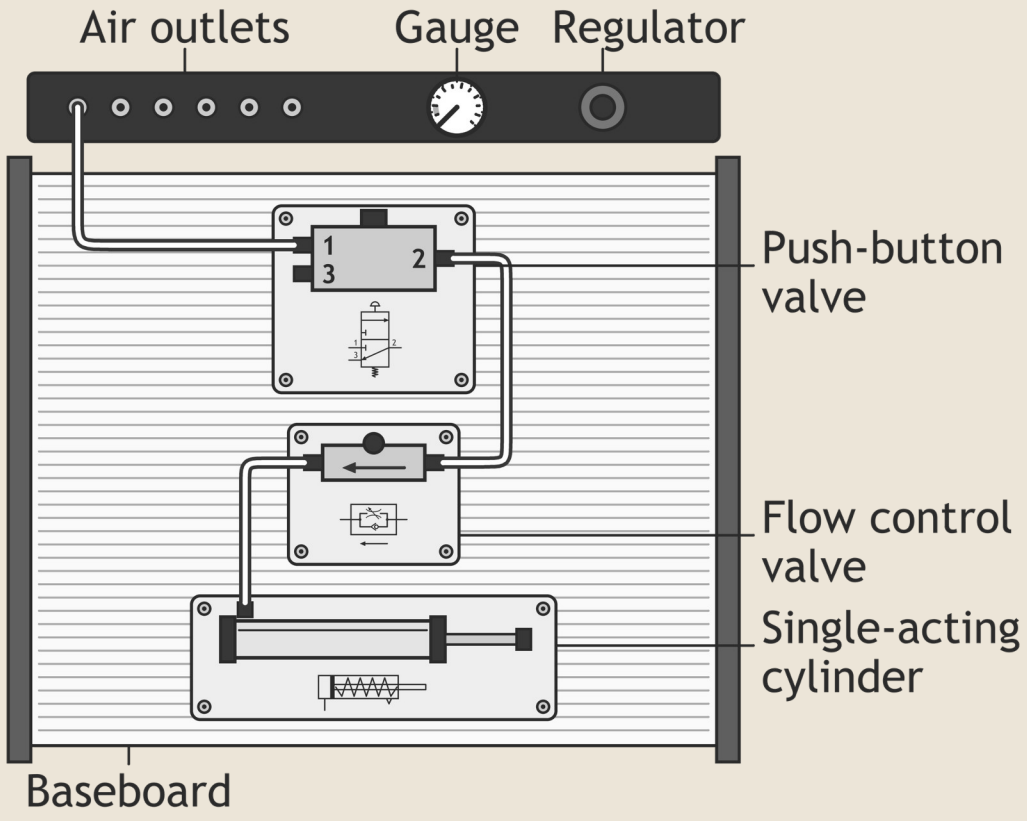
Slow the Gate Pin

Simulate it. Air Permit. Then the kit.

What to Collect

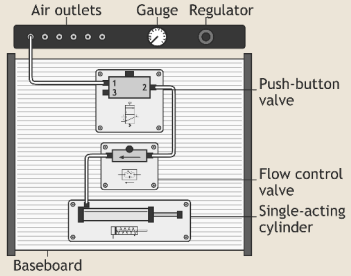
Builder and Safety: one set per team.

1 Bench card 3
2 Manifold (air supply)
3 3/2 push-button valve
4 Flow control valve
5 Single-acting cylinder
6 Tubing
7 Goggles, one pair each



Simulate and Build

Tasks 1-2 (B6A-L3-p7-8)

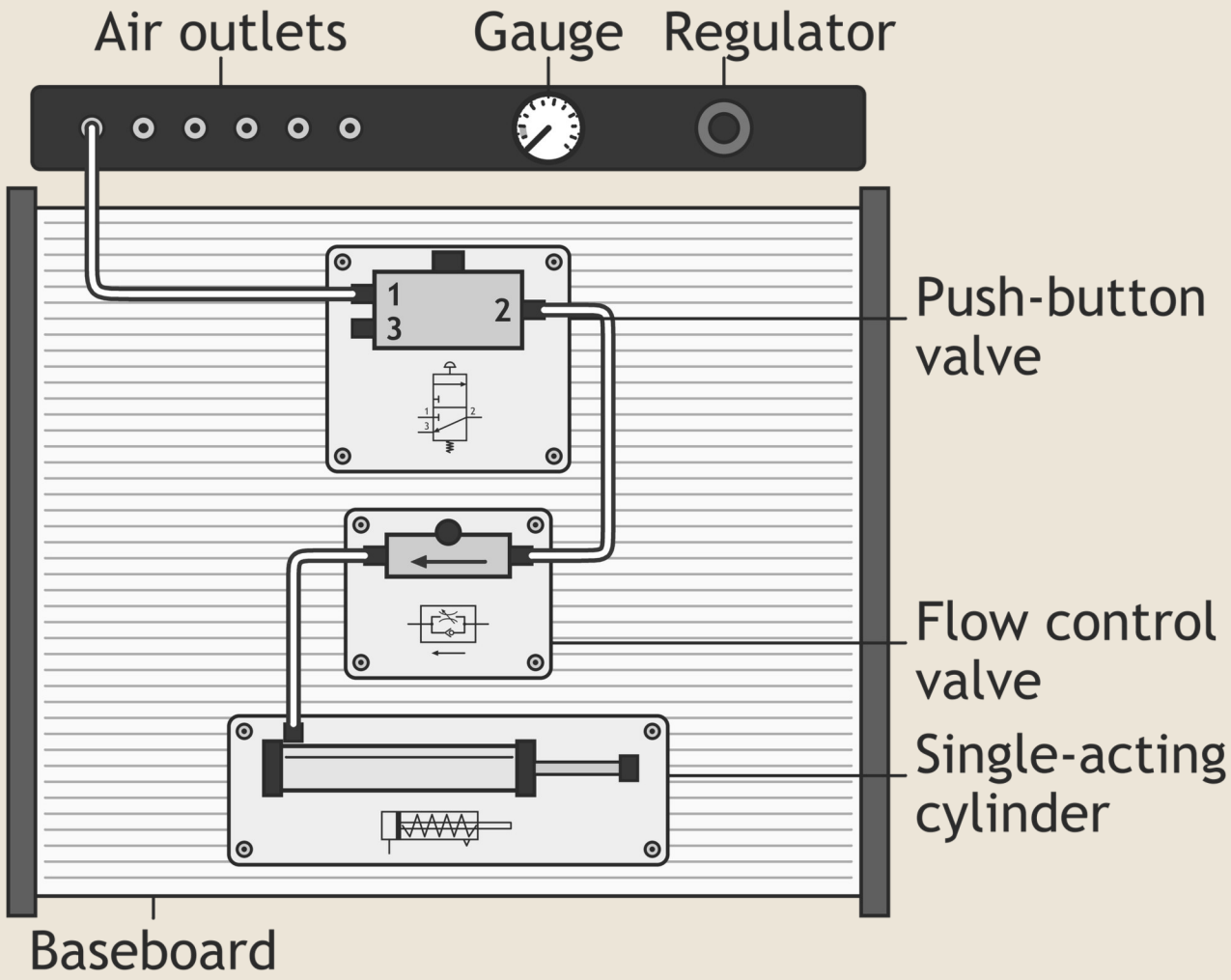


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 3.
Safety, then teacher: check the Air Permit.

SAFETY Turn the screw with air on, never a tube.

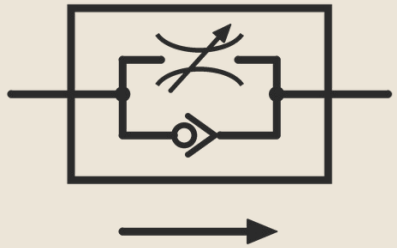
14 min

Build It: Bench Card 3



Find Out: Turn It Round

Task 5 (B6A-L3-p9)



Regulator to 0. Turn the valve round.
Safety checks every tube.
Test: press, let go. Fill in the table.

SAFETY Regulator at 0 before you turn the valve round.

5 min

Test It: What Did You Find?

1	Arrow to the cylinder: out… back…	Out slow, back fast
2	Arrow away from the cylinder: out… back…	Out fast, back slow
3	So the valve slows the air…	One way: the arrow's way
4	Points i and iii met? Which way round?	Yes: arrow to the cylinder

A Fair Test



Change one thing

the screw: right in, 1 turn, 2 turns, right out



Measure

the seconds for the rod to go out

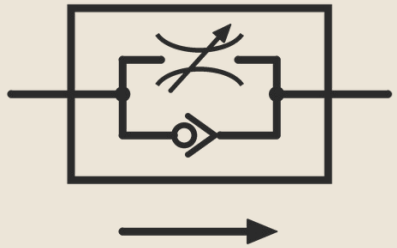


Keep the same

the pressure, the cylinder, the tubes, who counts

Fair Test: How Fast?

Task 3 (B6A-L3-p8)



Regulator to 0. Arrow back to the cylinder.
Point ii: time the rod at 4 screw settings.
Then Question 1: one thing kept the same.

SAFETY Fingers clear of the rod.

6 min

Beat the Clock (Optional)

Task 4 (B6A-L3-p8)



Set the pin to go out in 3 seconds.
Bronze 1-5 s, Silver 2-4 s, Gold 3 s (checked).

SAFETY Fingers clear of the rod.

4 min

Answers: Task 3 (B6A-L3-p8)

1. Q1 One thing kept the same

   The pressure (or the cylinder, tubes)

2. Point ii met? Screw in: the time…

   Goes up: yes

The Circuit You Built

Input

push-button valve

Control

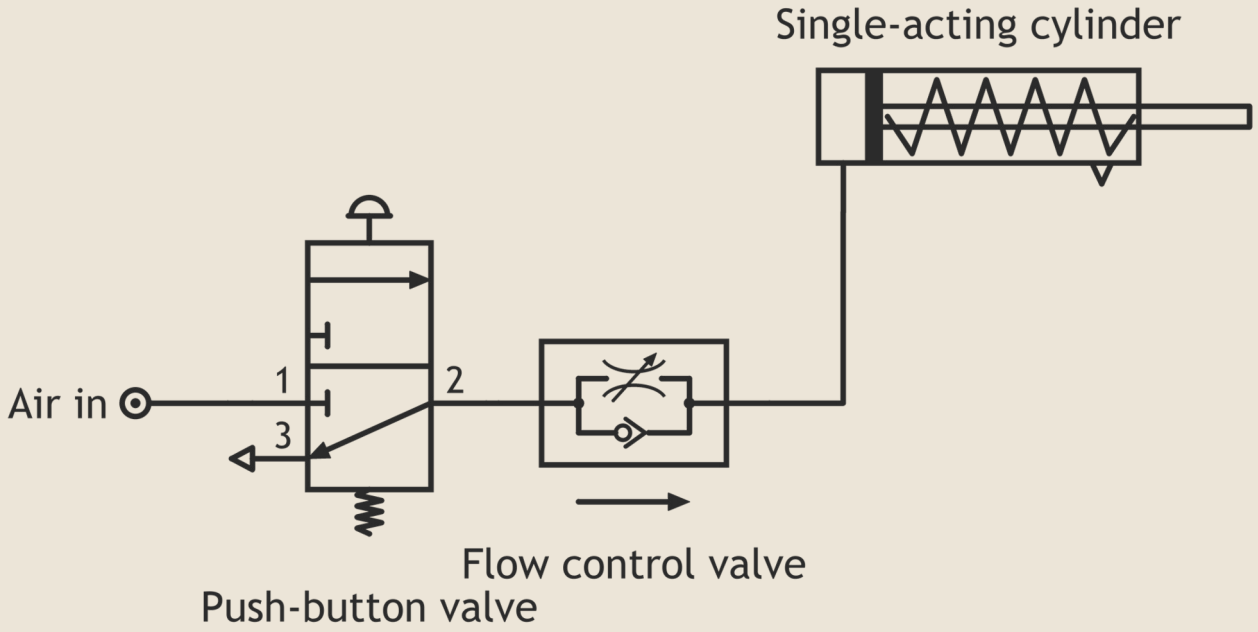
flow control valve

Arrow

to the cylinder

Output

the cylinder



The Flow Control Valve

Level Up

1 Screw

sets the size of the gap

2 Gap

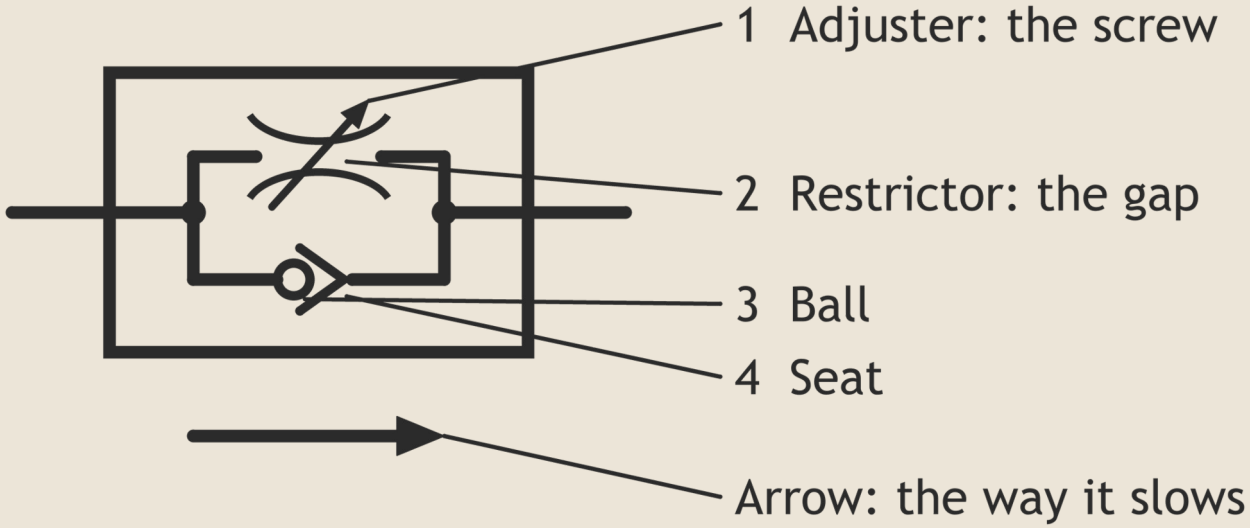
slows the air

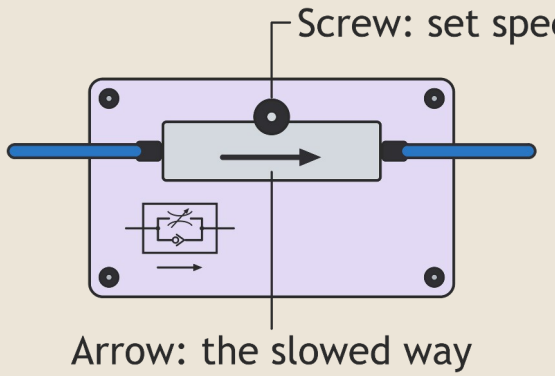
3, 4 Ball and seat

one way only

Arrow

the way it slows

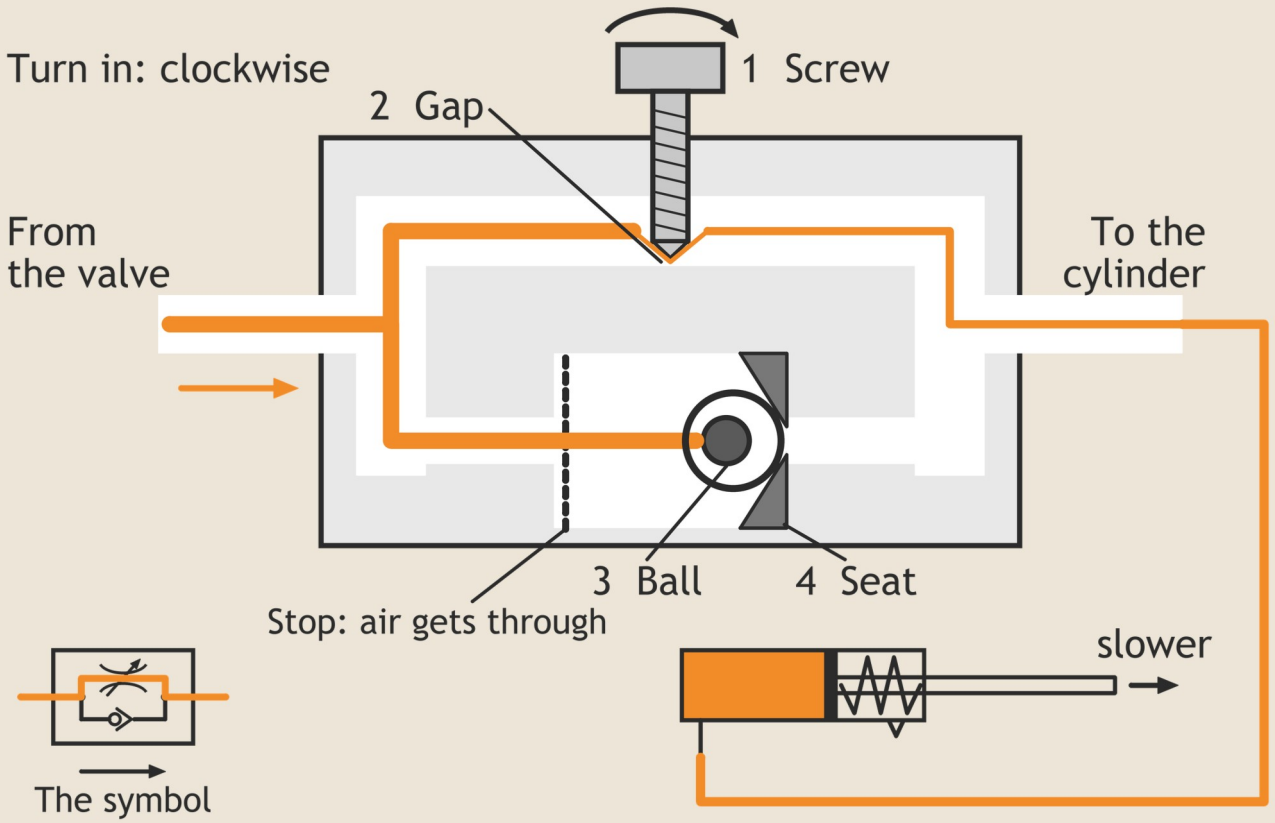




The real valve. Its symbol is on the plate.

Inside: the Arrow's Way

Level Up



1 Air arrives from the valve.

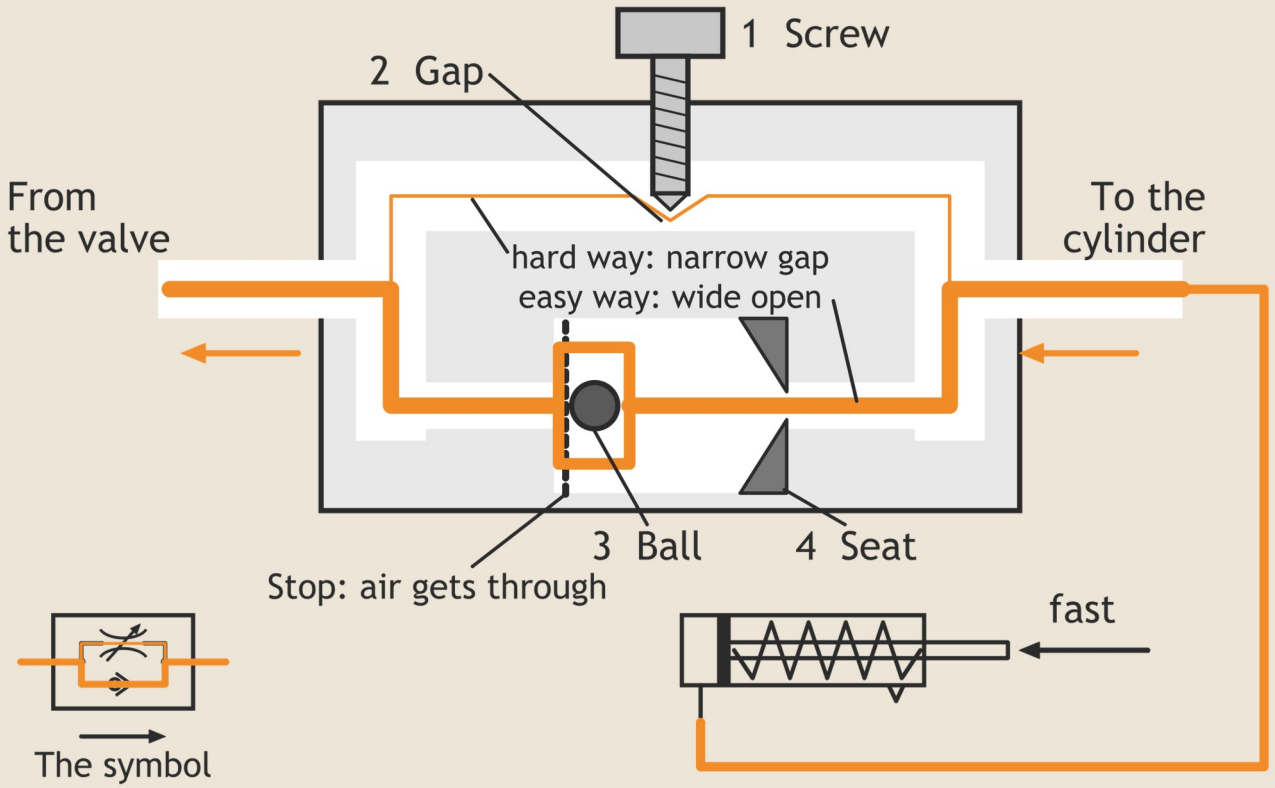
2 It pushes the ball onto its seat: shut.

3 It squeezes through the gap: slow.

4 Screw in: smaller gap, slower.

We Do: Coming Back

Level Up



1 Air comes back from the cylinder.

2 It pushes the ball off its seat.

3 Most air takes the easy way: fast.

The Exhaust Silencer

Level Up

Fits

in an exhaust port

Does

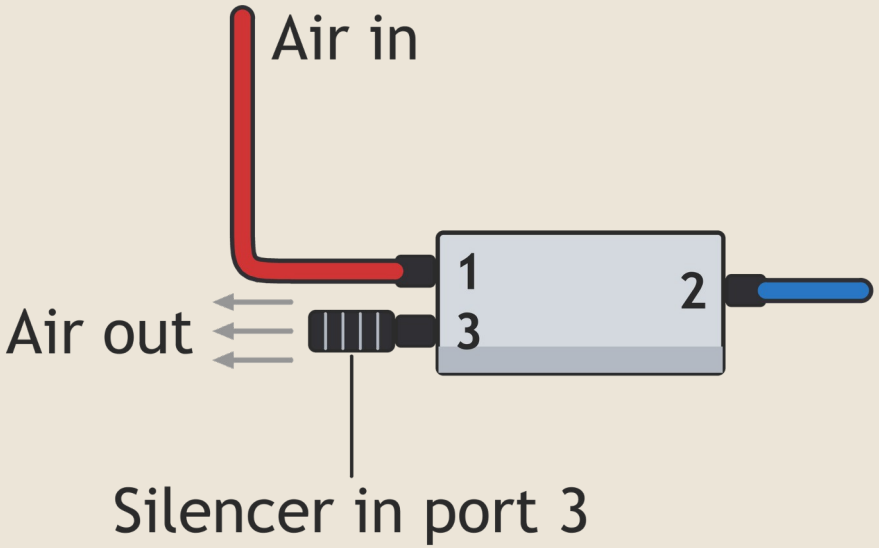
makes the air quieter

Has no

screw and no ball



The exhaust silencer symbol



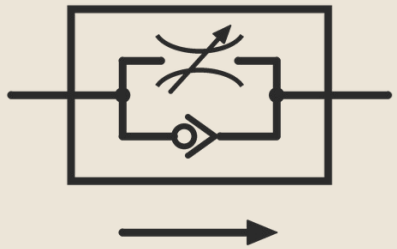
Questions, Key Words and Exam

Tasks 6-8 (B6A-L3-p9-10)

Task 6: Questions 2-3; draw Q4 in the box.

Task 7: Key words.

Task 8: draw the restrictor for the arrow shown.



7 min

Answers: Tasks 6-7 (B6A-L3-p9-10)

Question	Mark 1	Mark 2
Q2 (1 mark)	It slows the rod	-
Q3 (2 marks)	Ball pushed off seat	Air flows freely
Key words 1-2	slows	arrow
Key words 3-4	freely	restrictor

Answers: Task 8 (B6A-L3-p10)

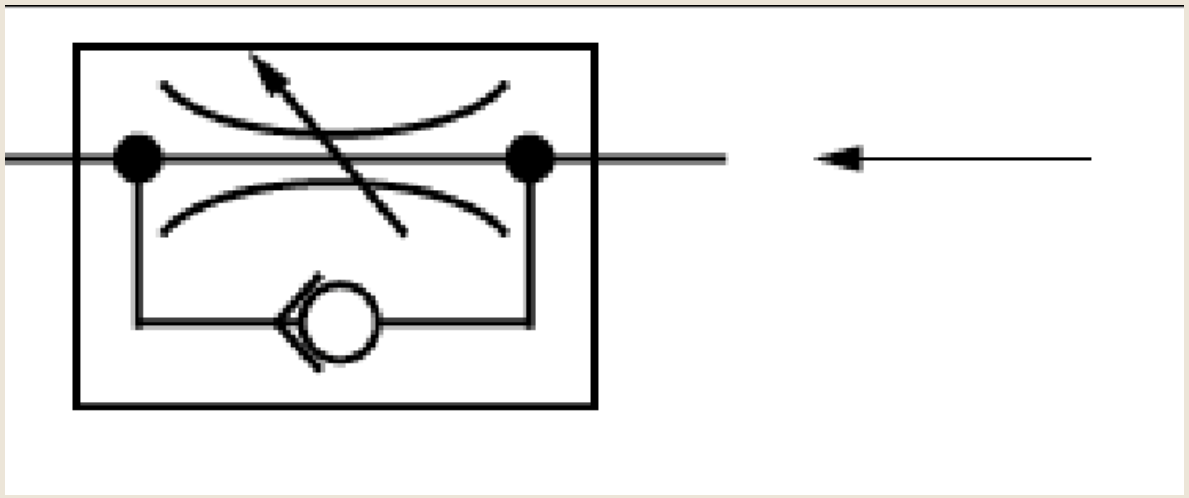
Symbol (1)

gap with adjuster, and ball and seat

Way round (1)

the V points the way the arrow points





Plenary: Control the Flow

1	What does a flow control valve slow?	The air, one way only
2	Turn the screw in: the rod goes out…	More slowly
3	Air coming back pushes the ball…	Off its seat
4	Arrow to the cylinder: the rod goes out…	Slowly
5	In a fair test you change…	One thing

Look back — by the end of this lesson…



1. I can slow a cylinder with a flow control valve
2. I can plan a fair test: keep the same, change one thing
3. I can say which way the arrow slows the air

LOOKING AHEAD

Lesson 4: Push and Pull

What if air pushes the rod out AND back in?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Flow control circuit: The Lesson 3 circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 2 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Flow control valve and circuit pictures: The Lesson 3 circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 2 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real valve drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Task 8 and Model Answer 2022 Q15(b): © Qualifications Scotland (SQA). N5 Engineering Science 2022, Question 15(b) and its marking instructions.

R Stewart - October 2026